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Improving lung cancer health equity by applying deep learning to low dose CT screening of minority and disadvantaged patients.

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Background: In the US, disparities in lung cancer mortality exist for African American, Hispanic, and other minorities. Standard of care low dose CT screening (LDCT) detects early-stage disease and improves mortality, yet disparities are perpetuated in screening by eligibility criteria derived from cohorts underrepresenting these minorities. One such cohort is the National Lung Screening Trial (NLST) cohort which is 92% White. Consequently, guidelines for lung cancer screening may be insufficient to address the unique needs of diverse populations. We hypothesize that Artificial Intelligence prediction of future lung cancer risk from an individual's LDCT can partially mitigate racial and ethnic disparities and improve health system practice guidelines by individualizing screening risk as compared to current general guidelines. Here, we benchmark a Resnet18 3D neural network trained on NLST LDCT images, Sybil, on the diverse patient population of the University of Illinois Health system (UIH) which is 20% White and 60% African American. **Methods:** A real-world cohort from UIH consisting of 1,450 CT studies was evaluated alongside 60,378 CT studies from the NLST cohort. All CT studies evaluated by the model were not used in model training. Using Youden's J index as a probability cutoff, sensitivity, specificity, positive predictive value, negative predictive value, and accuracy were evaluated. Receiver operating characteristic (ROC) and precision-recall (PR) curves were generated to assess model performance between cohorts. NLST data were truncated to achieve equivalent incidence of lung cancer with UIH when generated PR curves. **Results:** Multi-year prediction performance (ROC-AUC and PR-AUC) between cohorts are summarized (Table). For prediction of lung cancer within 1-year of LDCT in the UIH cohort, the model respectively demonstrated sensitivity, specificity, positive predictive value, and negative predictive value among White (0.80, 0.77, 0.54, 0.92), African American (0.84, 0.78, 0.39, 0.97) races and Hispanic (0.75, 0.73, 0.60, 0.84) and non-Hispanic (0.87, 0.77, 0.42, 0.97) ethnicities. **Conclusions:** Model performance was similar between the NLST (92% White) and a diverse, real-world cohort at UIH (20% White) though decreases in ROC-AUC performance in year 1 predictions and may be due to insufficient representation of minority populations during model training. Prospective studies involving larger and more representative patient populations should be conducted to further optimize the model and evaluate its clinical utility to improve lung cancer health equity in minority populations. Research Sponsor: None.

Model performance in NLST versus UIH cohorts.

Year diagnosed	NLST ROC-AUC	UIH ROC-AUC	NLST PR-AUC	UIH PR-AUC
Year 1	0.94	0.87	0.59	0.62
Year 2	0.86	0.85	0.53	0.68
Year 3	0.82	0.85	0.54	0.69
Year 4	0.79	0.84	0.57	0.70
Year 5	0.78	0.84	0.61	0.70
Year 6	0.77	0.83	0.63	0.70

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